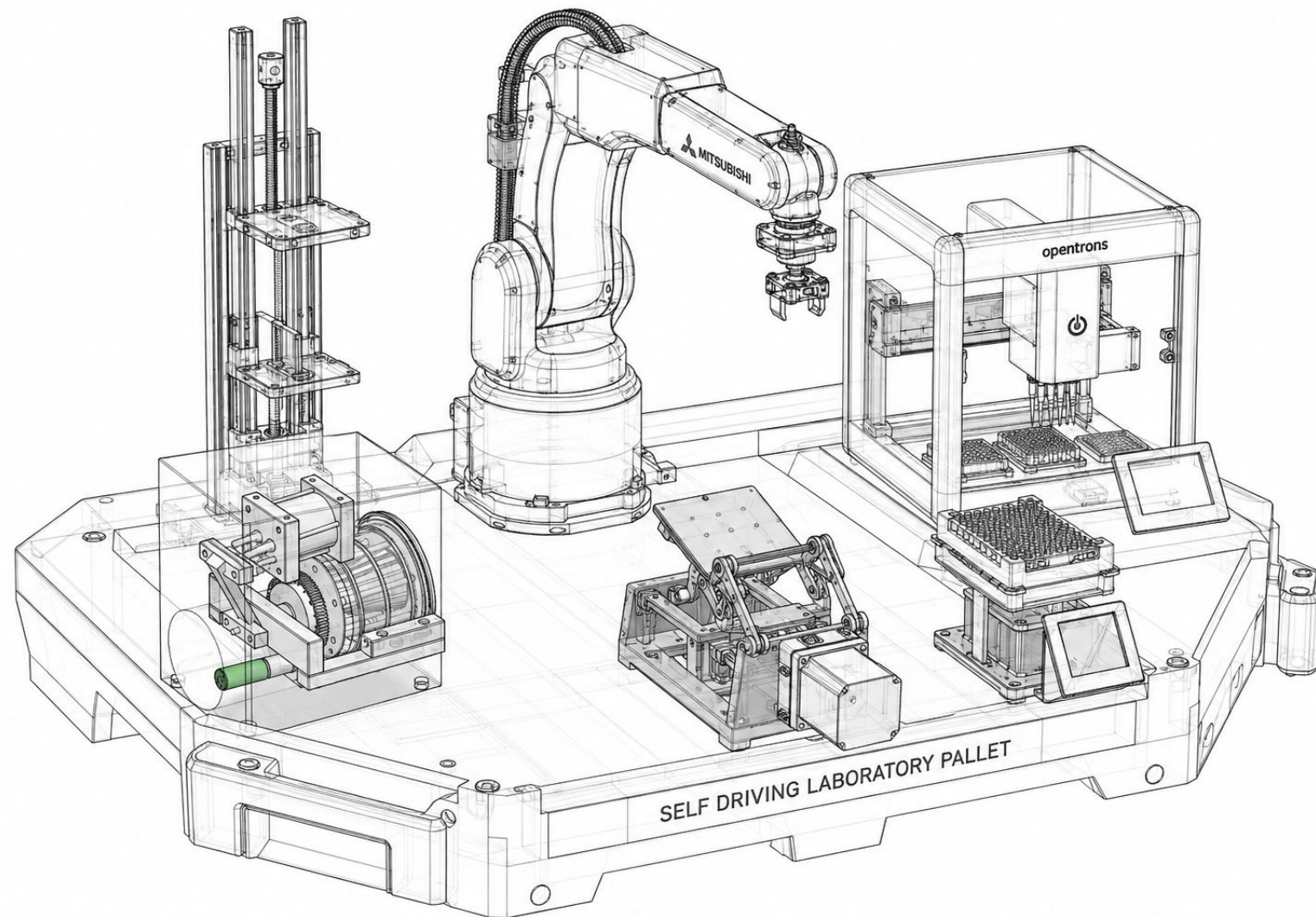


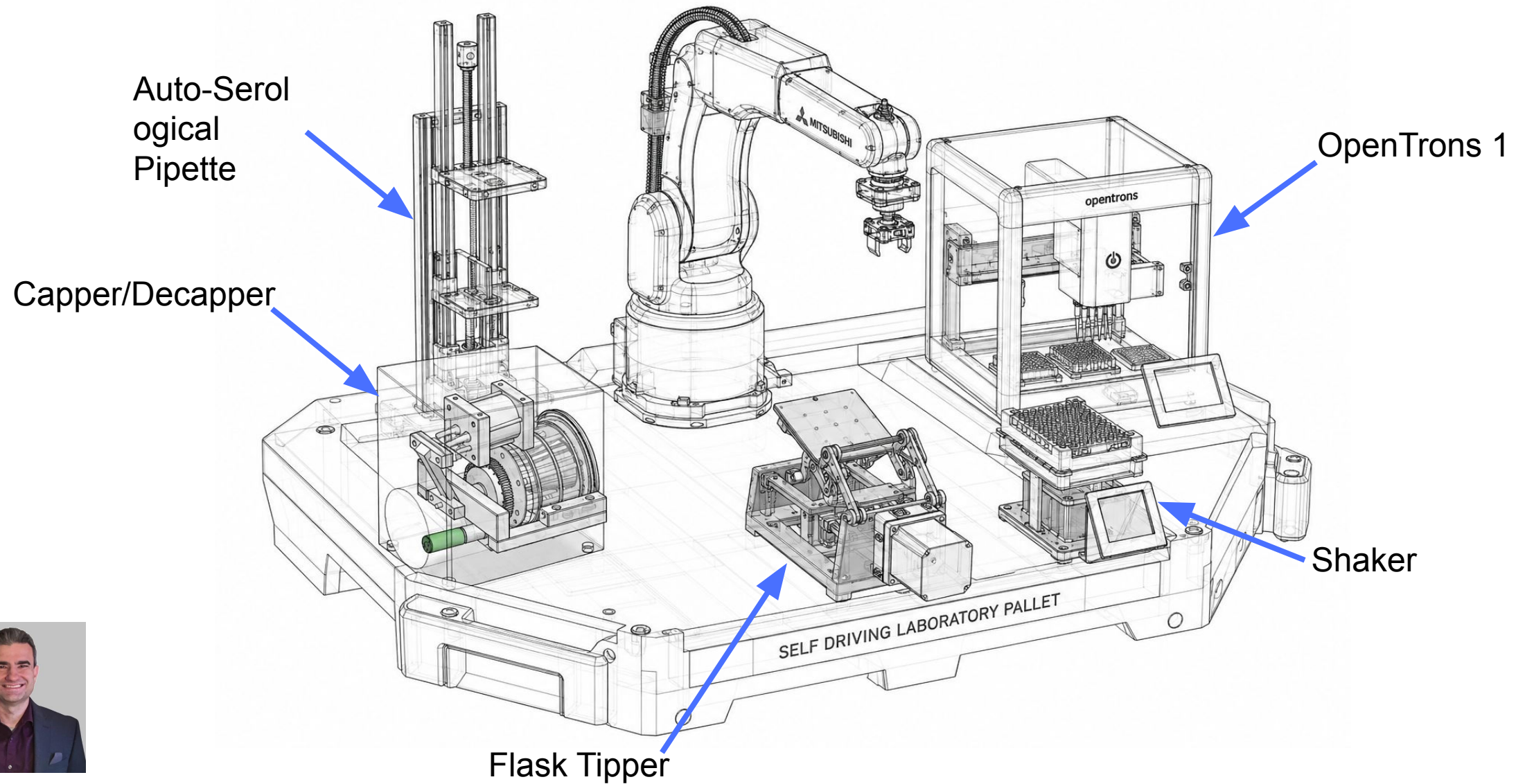


SDL

BUILDING THE LAB THAT EXPLORES WITH YOU

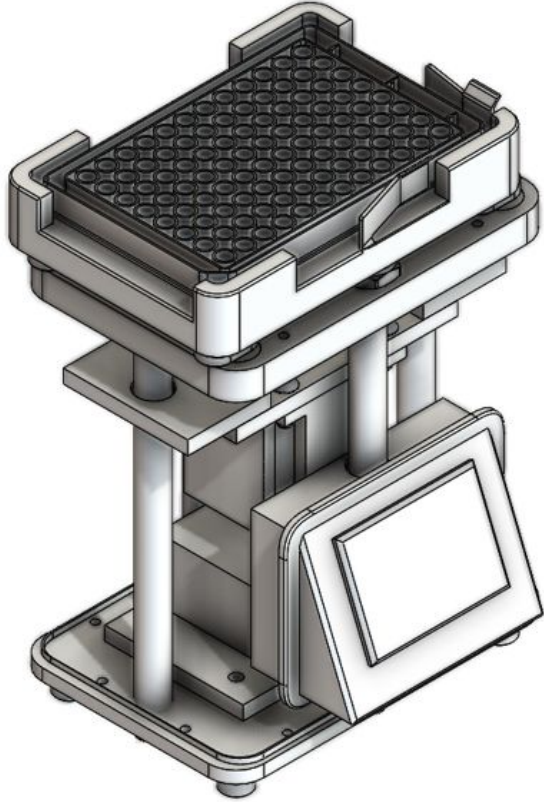
g. 1.

SDL Studio



DALE HERZOG

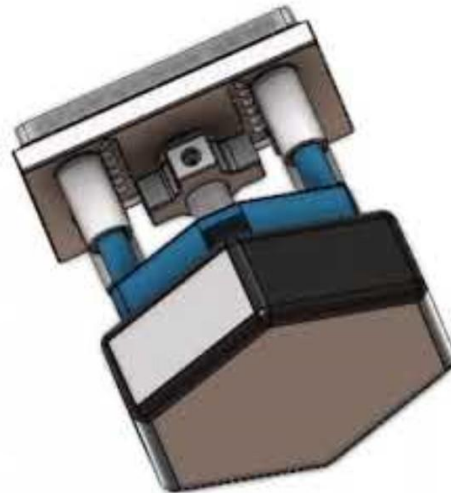
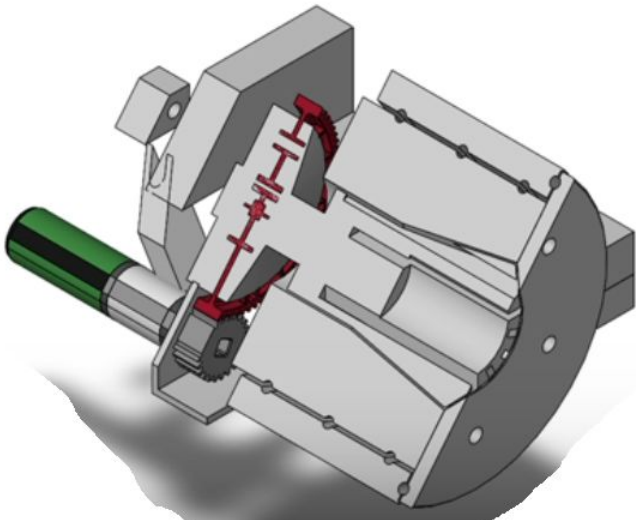
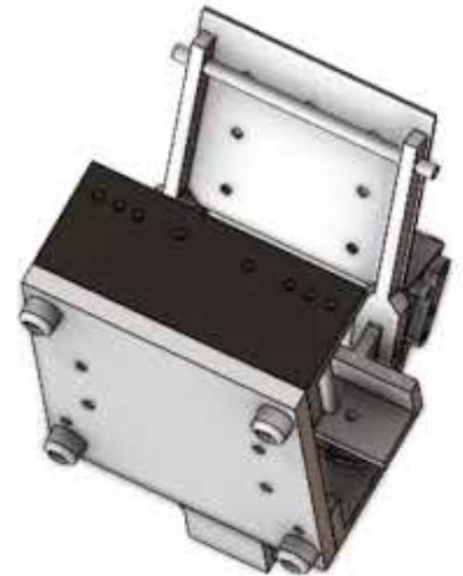
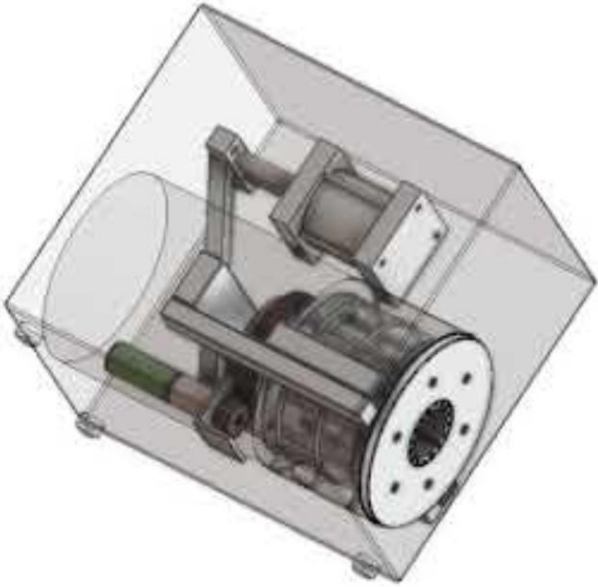
SDL Studio: Shaker



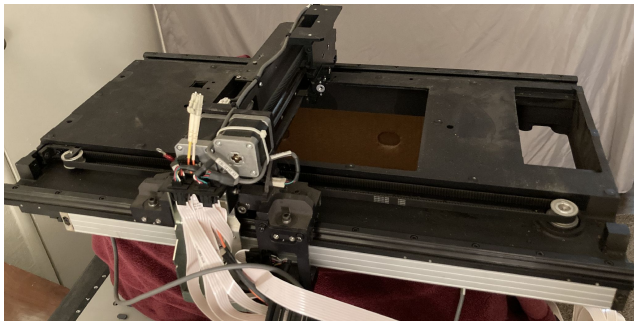
Shaker: Base Model



SDL Studio: Baseline Instrument Set



SDL Studio: Advanced Instrument Set

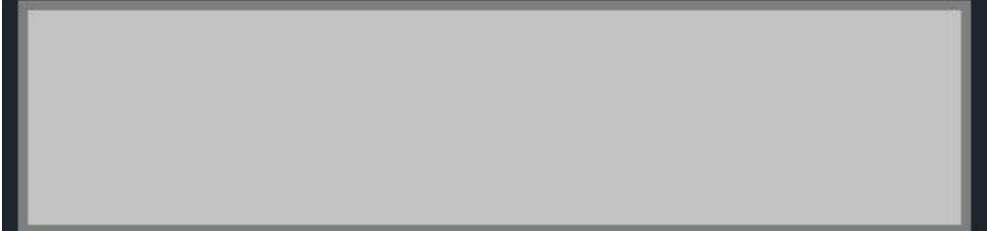


OT-1 Protocol Block Editor generates opentrons==2.5.2 Python — real API surface, real hardware limits unsaved changes

- Setup
- Deck & Labware
- Tips
- Liquid Handling
- Macros
- Movement
- Run Control
- Calibration
- Peripherals / Jog (advanced)

```
connect to robot
simulate only ☒
place tiprack-200ul in slot A1 as variable tiprack
place 96-flat in slot B1 as variable plate
place trough-12row in slot C1 as variable trough
add pipette
mount A (left mount) as variable p200
channels 1 (single)
min vol (uL) 0 max vol (uL) 200
aspirate speed (mm/min) 300 dispense speed (mm/min) 500
p200 pick up tip from well A1 of tiprack presses 3
p200 transfer 100 uL from well A1 of trough to well
p200 drop tip at well trash of trash-box home after ☒
home entire robot
```

GENERATED PYTHON



SDL Studio: Heater / Shaker: In Depth

Server Administration

Shaker Admin

Control panel

LINK

Web control

LINKED (web)

Last client

(none yet)

E-stop

clear

INSTRUMENT

Serial link

OK

Plate

Not loaded

Home

Not homed

RPM (current / target)

0 / 0

Protocol

(idle)

Plate check

enforced

ADMIN CONTROLS

These use the same endpoints as the client page and obey the same lock — they will refuse ON/run/jog while linked is false, same as the tablet. This is another view of the same control, not a bypass.

OFF

ON

- 5 RPM

0 RPM target

+ 5 RPM

Home

Paste a Shaker DSL script...

Run protocol

Seizes web control without waiting for Connect on the TFT. Refused only if E-stop is latched — there is no other local-control veto right now. That refusal is the one piece of the eventual production rule already wired in: once a physical E-stop is reinstalled, admin still cannot override local control while it's pressed.

E-STOP

Power-cycle Arduino

Power-cycle triggers an actual MCU reboot (DTR toggle) — not a graceful stop. Refused while a protocol is running. The board takes ~2s to come back; local/TFT state resets to Local on boot, and this page's "web control" flag is cleared to match. Real hardware only.

Local Control

Shaker

IDLE

remote

10

RPM

max 150

E-STOP

Options

Linked

Plate empty

Home not homed

DEV: NO PLATE

BENCH: NO E-STOP

Web Client Control

Shaker Control

connected

Help

Shaker Connected

Dev Mode: OFF

STATUS

State

idle

Instrument link

OK

Plate

Not loaded

Home

Not homed

Motor RPM

0

MANUAL CONTROL

OFF

ON

- 5 RPM

Target RPM

0

+ 5 RPM

Home

PROTOCOL

Shaker Practice Script

S(25) #Sets the rotation speed to 25 RPM

Shake(00:00:05) #Sets the shaker run time for 5 seconds

ShakerON #Runs the shaker for the previously defined run time and then automatically turns it off.

E-STOP

SDL Studio: Design To Request

Shaker Design Requests:

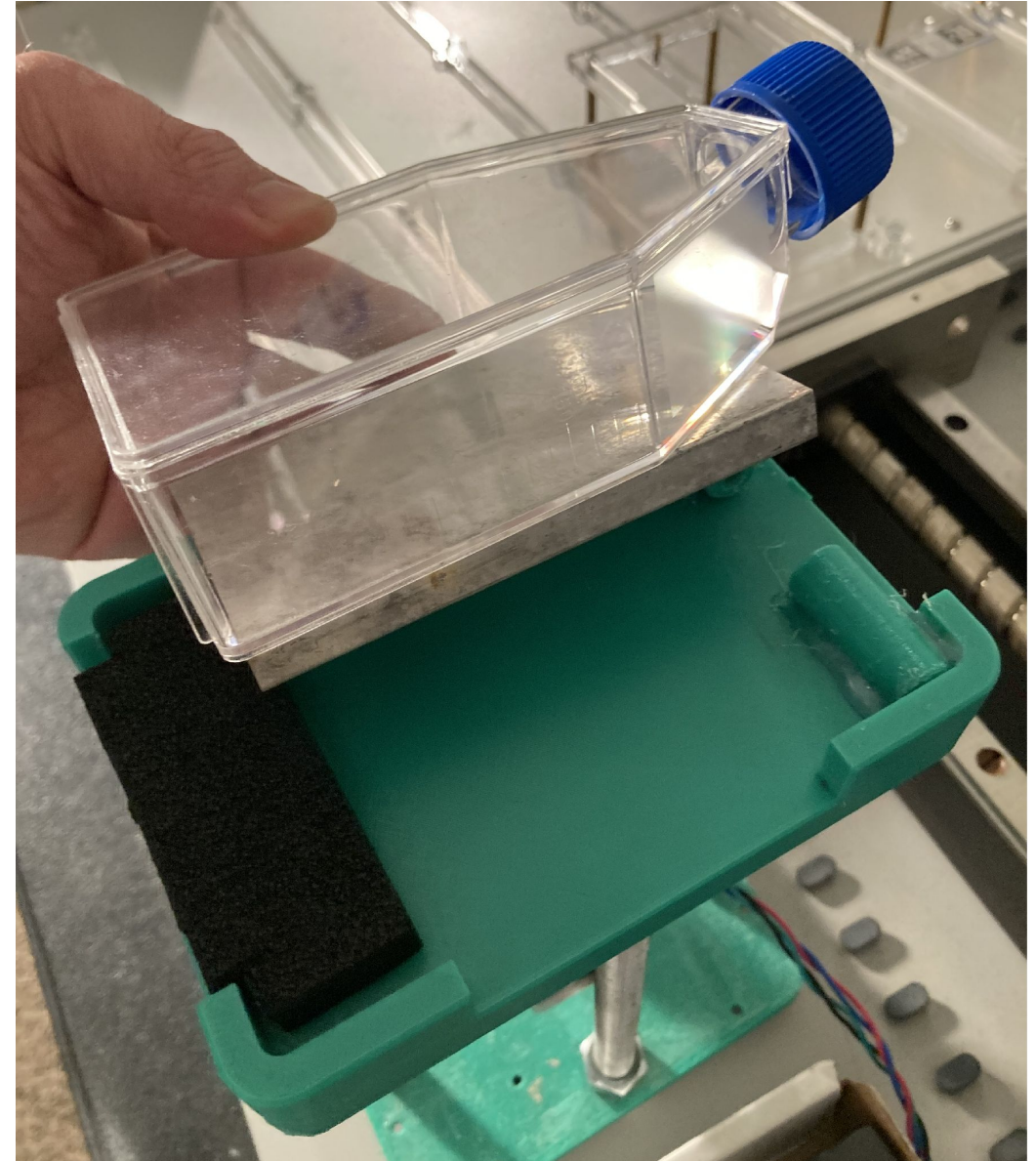
1. Modified Shaking Angle: **Designed and Built**
2. Precision Thermal Control: **Modeled**

Serological Pipette and Plate Handling Requests:

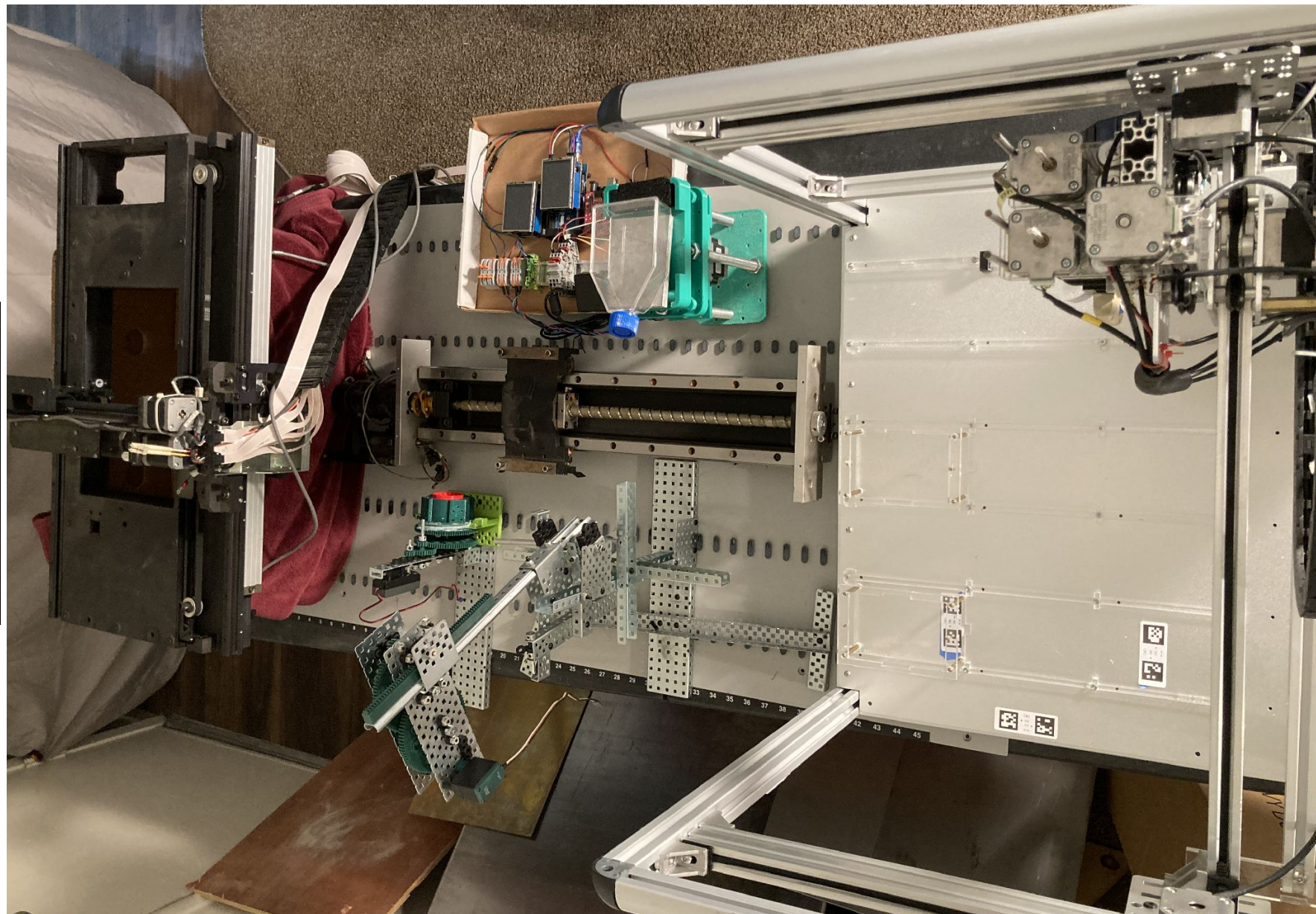
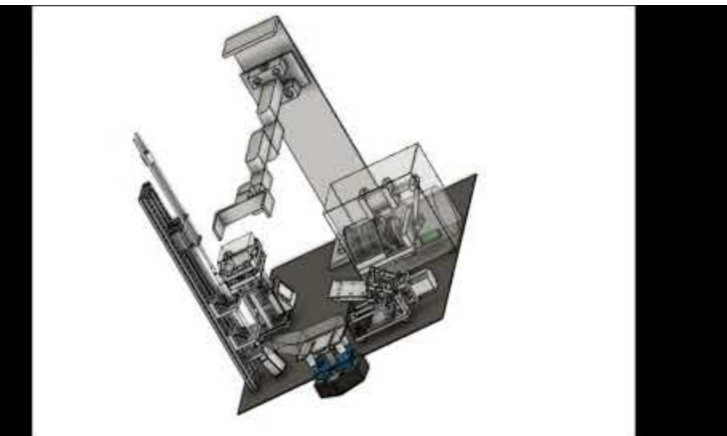
3. Simplify Iterative Process For Repetitive Workflows: **“Blockly” Design Interface**

General Equipment Automation Requests:

4. Robust Task Execution And Failure Recovery: **“Adaptive” Design In Progress Camera Tracking**
5. Automated Calibration: **Design In Progress Using Both Visual and Sensor Based Solutions**



SDL Studio: LAB 0



SDL Studio: 90 Day Plan

Month 01 –

Motion Control Design

1. Specify Components
2. Design & Simulate Code
3. Wire & Test Bare Components with Jetson Orin

Month 02 –

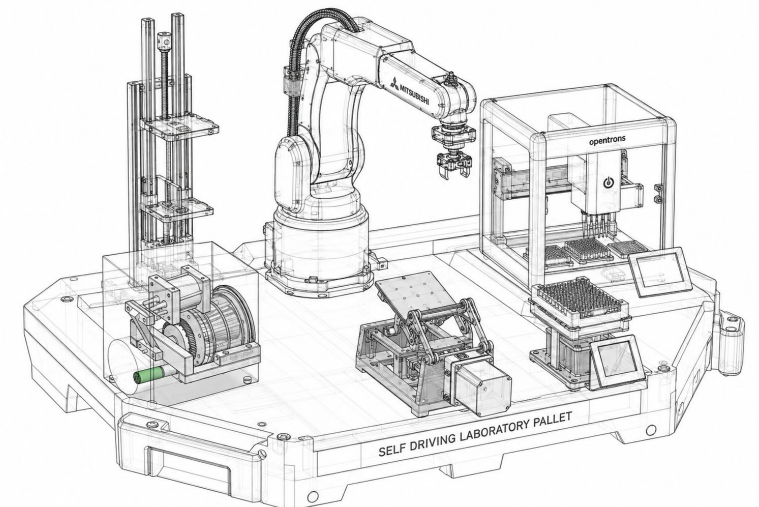
Sterile Design Update Software Update Design

1. Create Working Copies of Software, Setup, Orchestrator, Maintenance, + Integration with Data Platforms.
2. Build Washdown Compatible Instruments

Month 03 –

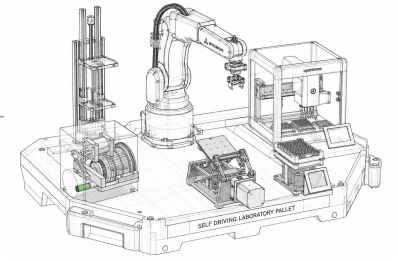
Integration & Testing

1. Automated Shakedown Testing
2. “Food Coloring” Shakedown



SDL Studio: Thank You

SPECIAL THANKS – WorldWide Studios .



Mentors & Co-Fellows

Fresno Ideaworks · Root Access Hackerspace · Klippenstein Corp, Counter Culture Studio 45 · Monomer Bio · Zeon Systems · Anthropic PBC · NVIDIA Corp.

Mentors & Co-Fellows of the WorldWide Studios AI Science Fellowship

Seeking Instrument Design and Lab Experience Fellowship: <https://www.linkedin.com/in/daleherzog/>

GUEST MENTORS



LUIS VILLA
LAB AUTOMATION ENGINEER



SIMON COELHO
SR. SOFTWARE ENGINEER



SRUSTI SAIN
AUTOMATION ENGINEER
@ L.A. SCIENCES



Michael Raspuzzi
Founder @ Worldwide Studios



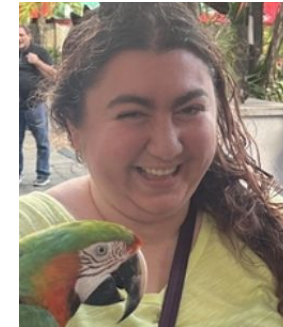
DYLAN HOLLAND
CO-FOUNDER @ WORLDWIDE
AI ENGINEER + SERIAL ENTREPRENEUR



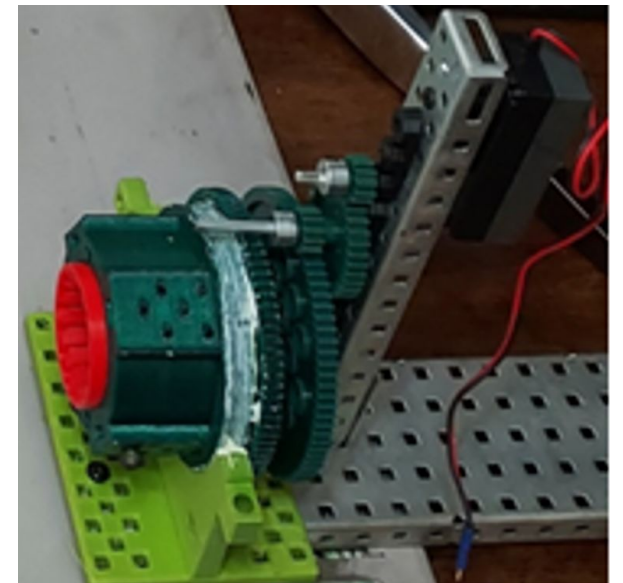
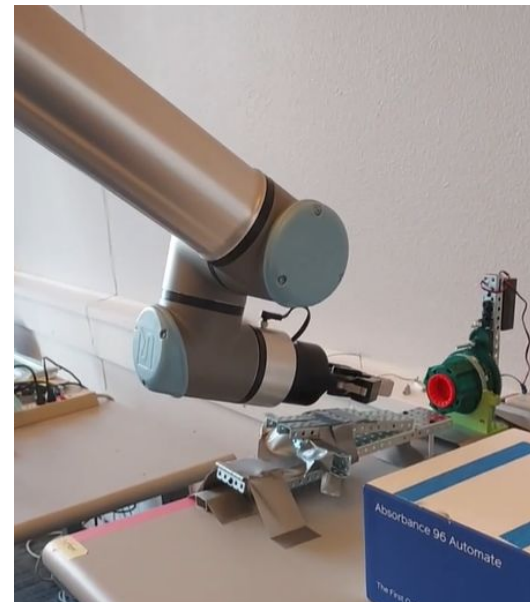
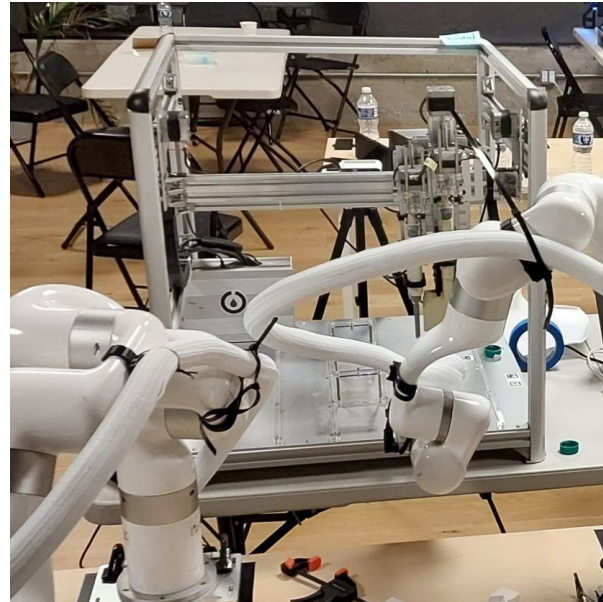
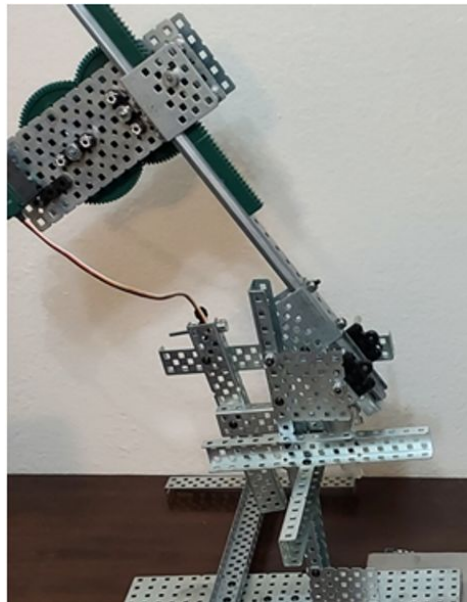
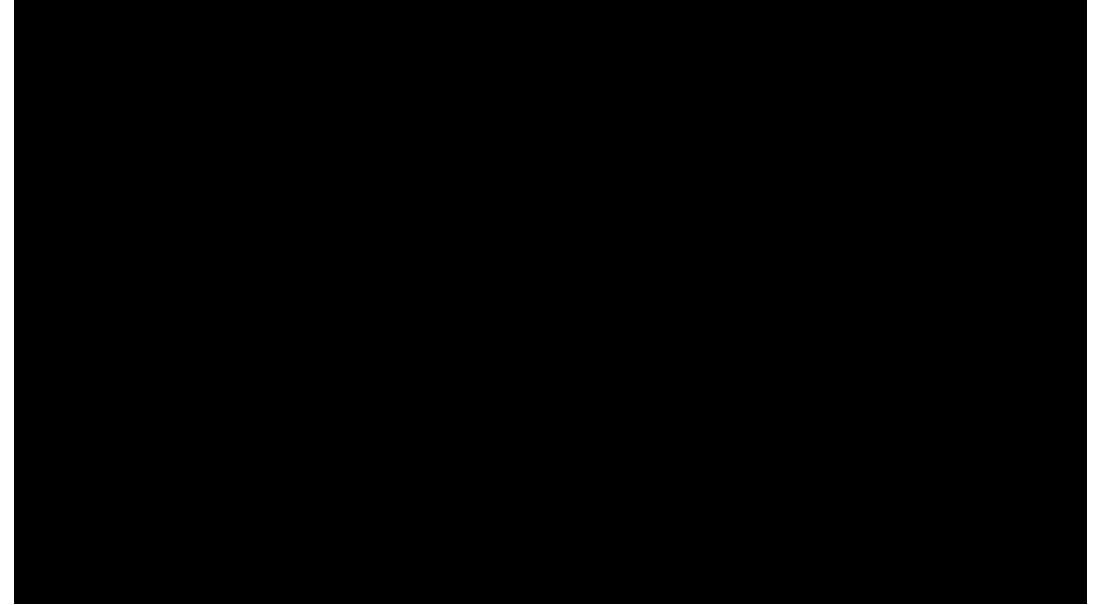
WORLDWIDE-STUDIOS.ORG
SPRING 2026
AI SCIENCE
GUEST MENTOR
NIKHIL DUBEY
ROBOTIC ENGINEER @ ANIMATICS



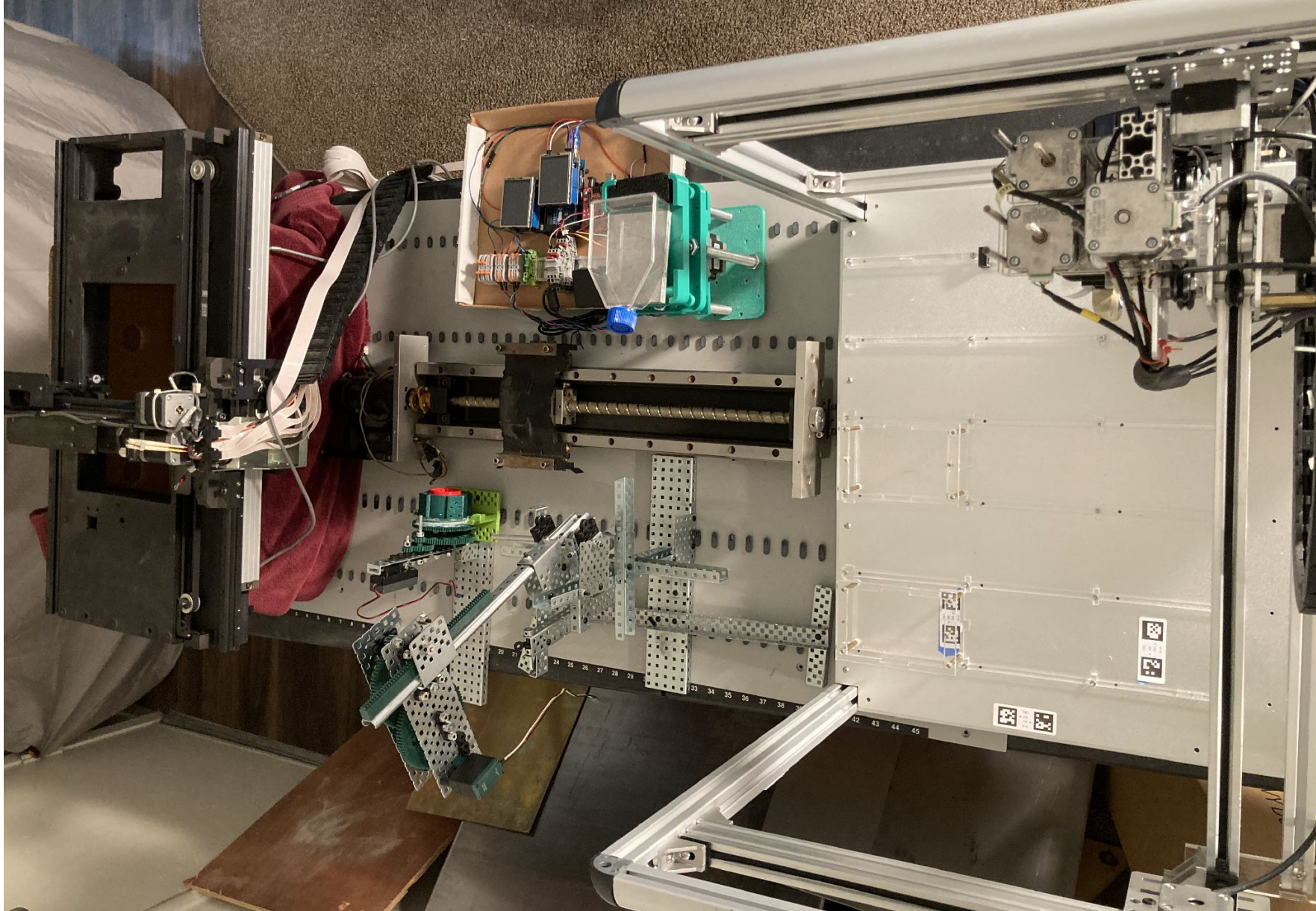
WORLDWIDE-STUDIOS.ORG
SPRING 2026
MENTOR
AI SCIENCE FELLOWS
ELVAN CAVAC
FOUNDING @ BETA



SDL Studio: Hackathons & Home Labs



SDL Studio: LAB 0



SDL Studio: Monomer Bio UR10 Arm Test

